The public leaderboard leaks the test labels, and the leak can be laundered invisibly

PRC Data Challenge 2026 · sent privately to the organisers by a participant · not for circulation before the challenge closes

The board scores every submission on all **N = 344,841** test rows and shows the RMSE to four decimals, with no private split. Every score is therefore an exact equation in the hidden labels. Below: how that is read out, how precisely, how it is turned into a clean-looking winning submission, and why only a hidden final evaluation stops it.

1. Every score is an exact equation

Take any scored file p_0 (score s_0) and submit $p = p_0 + d$ for a chosen change d. Then

$$\mathrm{N}(s^2 - {s_0}^2) = \Sigma\, d^2 + 2\,\Sigma\, d(p_0 - y)$$

Everything except the labels y is known, so each submission reveals the exact value of $\Sigma d(p_0 - y)$, a linear measurement of the test labels.

2. Reading single rows and groups

- **One row:** shift row i by Δ = 20,000 s. Its error is $(p_0 - y)_i = [\mathrm{N}(s^2 - {s_0}^2) - \Delta^2] / 2\Delta$, accurate to **±0.2 s** given four-decimal scores. Showing fewer decimals does not help: a larger Δ restores the precision.
- **Groups:** shifting k rows at once returns their summed error; a few nested group probes (a binary search) locate the anomalous rows in a large pool for a handful of submissions.
- **Why it pays here:** the squared error is dominated by a few hundred rows whose labels are hours to a day long (e.g. labels apparently measured from the scheduled rather than actual off-block time, and rows carrying an extra 86,400 s from a date rollover). No honest model can predict these; decoding a few dozen of them moves the RMSE by many points, and five submissions a day are plenty.

3. Ordinary submissions leak too, with no visible probe

For any set of scored files $p_\mathrm{k} = p_0 + d_\mathrm{k}$, the vector $T_\mathrm{k} = \Sigma d_\mathrm{k}(p_0 - y)$ is known exactly (section 1), and $G_\mathrm{kl} = \Sigma d_\mathrm{k} d_\mathrm{l}$ needs no labels.

The test-optimal blend $w = -G^{-1}T$ and its exact score follow by algebra. A team with dozens of diverse, perfectly normal-looking model submissions can extract several RMSE points this way. Its history looks like steady modelling progress, and nothing distinguishes it from "trying new ideas".

4. Laundering: the winning account stays clean

- **Separate the probing:** probes are submitted from a second account or by a collaborator. Their scores are absurd (tens to hundreds of RMSE above their best), but nothing links them to the competing team.
- **Hide the result in code:** the decoded rows and blend weights are absorbed by a model, e.g. added to training as "pseudo-labels", encoded as a "calibration table", or a model is simply fitted to reproduce the probed predictions. The published repository then genuinely produces the submitted file from open data.
- **Result:** clean code, a plausible score history, and a board score no honest model reaches. An AI or human code review cannot detect this, because nothing improper is in the code; the leak happened outside it.

5. What would fix it

- **A final evaluation on labels never scored on the board.** Best: a held-out period (e.g. further 2026 movements) on which teams submit predictions from their frozen pipeline shortly after the deadline. Decoded rows and board-fitted blends carry nothing over to new rows, so the ranking reflects the model.
- **Not sufficient:** splitting the current test set now (every row has already been scored in every submission), rounding the displayed score (larger shifts compensate), or code review alone (see section 4).
- **For future editions:** a public/private split with a small public share, a total submission cap, and a final round on unseen data announced from the start.

> **In short:** a full-test public board with four-decimal RMSE is an exact label oracle. Without a hidden final evaluation, the top of the ranking measures how well teams exploit and conceal it, and the method can be made undetectable. A short second round on unseen data is the only reliable remedy.